

Math 104– Lecture 01

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1 START

Definition 1.1 (Sequence). A sequence is a map from $n \in \mathbb{Z} | n \geq m (m \in \mathbb{Z})$ to a set X

Definition 1.2 (Convergence). A sequence $(s_n)_1^\infty \subset \mathbb{R}$ converges to $s \in \mathbb{R}$ if

$$\forall \epsilon > 0, \exists N > 0 \text{ such that } n > N \implies |s_n - s| < \epsilon$$

Definition 1.3 (Divergence). A sequence (s_n) diverges if (s_n) doesn't converge to any real number.

Definition 1.4 (Limit Notation). We write

$$\lim_{n \rightarrow \infty} (s_n) = s, \lim s_n = s, s_n^{n \rightarrow \infty} \rightarrow s$$

We write

$$\lim_{n \rightarrow \infty} (s_n) = \infty$$

if

$$\forall m \in \mathbb{R} \exists N > 0 \text{ such that } n > N \implies s_n > m$$

Example 1.5 (Convergence Example). Take $s_n = \frac{3n+1}{7n+4}, n \in \mathbb{N}$. We have

$$\begin{aligned} \lim_{n \rightarrow \infty} (s_n) &= \lim_{n \rightarrow \infty} \frac{3n+1}{7n+4} \\ &= \lim_{n \rightarrow \infty} \frac{3 + \frac{1}{n}}{7 + \frac{4}{n}} \\ &= \frac{3}{7} \end{aligned}$$

How do we verify this?

Proof. Epsilon Delta: Let $\epsilon > 0$. Take $N = \frac{19}{49\epsilon} + \frac{4}{7}$. if $n > N \rightarrow |s_n - \frac{3}{7}| < \epsilon$

The real step is finding N with equation

$$\left| \frac{3n+1}{7n+4} - \frac{3}{7} \right| < \epsilon$$

□

Theorem 1.6 (Uniqueness of the limit) *"Any sequence has a unique limit"*

Let $(s_n) \subset \mathbb{R}$. If $\lim_{n \rightarrow \infty} s_n = s \in \mathbb{R}$ and $\lim_{n \rightarrow \infty} s_n = t \in \mathbb{R}$ then $s = t$

Proof. Establish

$$\lim s_n = s \rightarrow \exists N_1 > 0 \text{ such that } n > N_1 \implies |s_n - s| < \frac{\epsilon}{2}$$

Establish again

$$\lim s_n = s \rightarrow \exists N_2 > 0 \text{ such that } n > N_2 \implies |s_n - s| < \frac{\epsilon}{2}$$

Then

$$|s - t| \leq |s - s_n| + |s_n - t| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Take $N = \max N_1, N_2$ Note that $N > N_1$ and $N > N_2$

□

2 Main idea

Write the one-sentence point of today's lecture here.

3 Definitions and notation

Definition 3.1 (First definition). Write the definition and one sentence explaining why it matters.

4 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

5 Examples and calculations

Example 5.1 (Lecture example). Write the setup, the calculation, and the conclusion.

6 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: